Khang B. Tran

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SUMMARY

Third-year B.S. Computer Science student aspiring to become an Application Developer, with hand-on experience building a Phyton-based route-optimization web application using Streamlit, OSMnx, and Folium. Familiar with Python, C++, SQL, JavaScript, and HTML/CSS, with experience developing regression test cases and validating a live web application as a Quality Assurance Intern. Interested in developing reliable, data-driven software and contributing to application design, development, testing, and implementation.

EDUCATION

City College of Angeles Angeles, Pampanga
B.S. in Computer Science *2024 – 2028 (expected)*

- GPA: 1.75 equivalent to 3.25 / 4.0
- Relevant Coursework: Data Structures & Algorithms, Computer Programming, Object-Oriented Programming, Information Management (SQL)
- Practice SELECT, JOIN, GROUP BY, subqueries, and basic database design

EXPERIENCE

Investagrams Inc. *Sep 2026 – Present*
Quality Assurance Intern (Voluntary Internship) | *Remote*

- Executed 426 test cases (142 × 3 environments) on the InvestaTrade web app with a 100% pass rate.
- Built a 142-case regression suite across 13 modules, including KYC, portfolio, Explore, watchlist and cash-in.
- Prioritized cases by risk: 71 High, 59 Medium, 12 Low.
- Wrote 21 Explore and 17 stock-detail cases (about 27% of the suite), plus 12 KYC validation cases covering boundary and negative inputs.

PROJECTS

Garbage Route Optimizer (AL-1) *Mar 2026 – May 2026*
Python Developer & Researcher

- Built a Streamlit web app that solves capacity-aware vehicle routing for Angeles City garbage trucks, using a Genetic Algorithm with 2-opt local search. Result:
- Used OpenStreetMap road networks (OSMnx) for real path distances, with a Haversine fallback to keep routing and scheduling consistent.
- Built interactive multi-truck maps (Folium), live convergence charts, and fuel, cost and CO2 tracking.

Network Design & Configuration (Cisco Packet Tracer)

- Designed a multi-router LAN with switches and end devices, applying IPv4 subnetting across separate networks.
- Configured VLANs 30 and 40, DHCP, multilayer inter-VLAN routing, and static routes between routers.
- Tested end-to-end host connectivity with ping and troubleshot IP addressing and routing through the Cisco IOS CLI.

IT Help Desk & Asset Management System | Java, Spring Boot, MySQL
Personal Project

- Built a full-stack IT help desk and asset management system using Java, Spring Boot, Thymeleaf, and MySQL.
- Implemented BCrypt authentication, role-based access control, ticket tracking, asset management, search, and dashboard metrics.
- Designed normalized database tables with Spring Data JPA and Flyway migrations.
- Added server-side validation, CSRF protection, business rules, and JUnit/Mockito tests.

ACTIVITIES AND LEADERSHIP

Republic Central Colleges Documentation Team *Aug 2021 – June 2023*
Business Manager / Videographer / Editor

- Produced and edited 15 school videos in Adobe Premiere Pro, reaching 41,575 total views.
- Ran the team's social media strategy, growing the school's Facebook following by 2,000.

SKILLS

Programming: Python, Java, C++, SQL, JavaScript, HTML/CSS
Frameworks/Libraries: Spring Boot, Spring Data JPA, Spring Security, Streamlit, Folium, OSMnx
Databases: MySQL
Testing & Tools: JUnit, Mockito, Pytest, Git, Visual Studio Code